

# Multi-Stage and Multi-Target Data Selection for Medical Foundation Models

Literature Review and 2026 Research Findings

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**Project:** Multi-Stage and Multi-Target Data Selection Harness for Medical Foundation Models  
**Status:** Working document. Sections 3 (CPT) and 7 (data selection) are the seed; dataset sections are for their owners to extend. See [contributing.md](#) for who owns what.

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## 1. Scope and how to read this

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This review supports the proposal's month 1-2 deliverable ("architecture document"). It covers four things, in descending order of depth:

- **Continual pretraining (CPT):** the stage the project is starting with, and the one where corpus selection plausibly matters most.
- **Data selection:** the methods this project intends to extend to a multi-stage setting.
- **The four datasets** we have committed to: MedQA, MedMCQA, PubMedQA, and the PubMed corpus.
- **SFT and preference optimisation:** enough to know what the later stages will need.

Three conventions. Claims that are load-bearing for a design decision carry a citation. Where a number was taken from a secondary source rather than read out of the primary paper, it is marked (*secondary*). Those should be checked against the paper before appearing in a manuscript.

Third: material added by the July 2026 research pass went through adversarial verification, three independent attempts to refute each claim, with a majority refutation killing it. Nine of 25 claims were killed that way and are not stated as fact here. Where a killed claim is nonetheless worth knowing about, because a reader might otherwise find the paper and draw it themselves, it is named explicitly as not surviving verification rather than quietly omitted. Two citations still carry an explicit caveat that a human has not yet read the primary source: [He26] and [Ya26], the two papers that most reshape Sections 8 Gap 1 and Gap 2. Read those before defending this document.

A caution that shapes everything below: **the field's headline claim is contested**. Section 3.5 covers evidence that medical domain adaptation frequently fails to beat the base model under fair comparison. Read that before designing experiments, not after.

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## 2. The multi-stage training pipeline

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Modern medical foundation models are built in stages, and the stages differ in kind, not just in hyperparameters:

| Stage                        | Objective                    | Data                       | What it is supposed to add                    |
|------------------------------|------------------------------|----------------------------|-----------------------------------------------|
| Continual pretraining (CPT)  | Next-token prediction        | Raw biomedical text        | Domain knowledge, vocabulary, long-tail facts |
| Supervised fine-tuning (SFT) | Next-token on responses only | Instruction-response pairs | Task format, instruction following            |
| Preference optimisation      | DPO / RLHF objective         | Preference pairs           | Helpfulness, safety, calibration              |

The important structural point for this project is that **each stage consumes a different kind of data and is credited with a different capability**. A corpus that is excellent for CPT (broad, diverse, long-tail) is not an instruction dataset, and an instruction dataset that teaches format does not teach facts. It follows that a single data-selection policy optimised once cannot be right for all three, which is precisely the proposal's thesis.

What the literature actually does, though, is optimise one stage at a time and evaluate on one metric, usually downstream accuracy on a single benchmark. Section 8 returns to this.

## 3. Stage 1: Continual pretraining

### 3.1 The founding result

The canonical reference is Gururangan et al. (2020), *Don't Stop Pretraining* [G20]. It established two now-standard moves:

- **DAPT** (domain-adaptive pretraining): continue pretraining on a large unlabelled corpus from the target domain.
- **TAPT** (task-adaptive pretraining): continue pretraining on the (unlabelled) text of the task itself, which is much smaller but far better targeted.

The paper reported gains across four domains including biomedical, and found DAPT and TAPT to be complementary. Crucially for us, it also showed that *selecting* task-relevant text from a large corpus (a cheap embedding-based retrieval) recovered much of TAPT's benefit. That is a data selection result, published in 2020, and it is the ancestor of the work in Section 7.

### 3.2 The recipe question: how to continue pretraining without destroying the model

The practical difficulty in CPT is that naively resuming training on a new distribution degrades prior capability: catastrophic forgetting. Ibrahim et al. (2024) [I24] give the clearest recipe result: a combination of

1. **learning-rate re-warming**,
2. **re-decaying** the schedule over the new run, and
3. **replay** of a fraction of the original pretraining distribution

is sufficient to match full retraining from scratch on the union of data, at a fraction of the compute. This matters to us in a specific way: **replay ratio is a data-selection decision**. Deciding what fraction of general-domain text to mix into a medical CPT run, and which general-domain text, is exactly the kind of thing our harness should be able to express.

Related practical findings worth carrying into experiments:

- The *ordering* of a CPT corpus is not neutral; curriculum-like effects exist, though the evidence is weaker than for replay.

- Sequence packing (concatenating documents into fixed-length blocks) is standard for efficiency. It introduces a subtlety we implement explicitly: without an EOS separator between documents, the model learns to continue one abstract into an unrelated one.

### 3.3 Medical CPT models

The main open-weight lineage, roughly chronologically:

- **BioMedLM** [B24]: 2.7B, trained from scratch on PubMed abstracts and full text rather than adapted. Useful as the "pure biomedical" reference point.
- **PMC-LLaMA** [W23]: LLaMA continually pretrained on biomedical papers and textbooks, then medically instruction-tuned. Reported strong benchmark results at much smaller scale than the general models of its day; explicitly frames itself as data-centric knowledge injection.
- **MEDITRON** [C23]: 7B and 70B, CPT on a curated corpus (PubMed Central full text, abstracts, clinical guidelines). Notable for treating corpus construction as the contribution.
- **BioMistral** [L24b]: Mistral 7B further pretrained on PubMed Central open access.
- **Me-LLaMA** [X24c]: LLaMA 2 with continued pretraining plus instruction tuning on a mixed biomedical and clinical corpus.
- **MedGemma** [M25]: Gemma 3 derivative (4B multimodal, 27B text and multimodal), pretrained on medical text, medical QA, and FHIR-based EHR data, with a SigLIP encoder pretrained on de-identified medical images for the multimodal variants. Reported to beat its Gemma base on all tested text-only health benchmarks.

For our purposes MedGemma is best used as a **reference ceiling**, not as a starting point: it is already medically adapted, so further CPT on it would make any improvement impossible to attribute to our selection method. This is why the harness targets Gemma 3 base.

### 3.4 What CPT is supposed to buy, and the measurement problem

The claim implicit in all of the above is that reading more biomedical text makes a model better at medical tasks. Measuring that is harder than it sounds:

- **Benchmark contamination.** PubMedQA is derived from PubMed abstracts. Continually pretraining on PubMed and then evaluating on PubMedQA risks training on the evaluation contexts. This is a live concern for our own experiments and must be handled by PMID-level exclusion.
- **Prompt sensitivity.** MCQ accuracy for base models varies enormously with prompt format and scoring convention. A "gain" can be an artefact of the adapted model happening to suit the prompt better.
- **Capability collapse.** Aggregate accuracy can rise while performance on rare or underrepresented categories falls. The proposal's capability-retention framing exists precisely because a single number hides this.

### 3.5 The contrarian result, and why it matters most

Jeong et al. (2024), *The Limited Impact of Medical Adaptation of Large Language and Vision-Language Models* [J24], evaluated ten medical LLMs and two medical VLMs against their own base models, optimising prompts **separately for each model** and accounting for statistical uncertainty. Their finding: on clinical-note QA with 3-shot prompting, the medical models beat their bases in only 26.7% of comparisons, tied in 16.7%, and *lost* in 56.7% (*secondary: figures taken from the paper abstract via its arXiv page*).

Their diagnosis of prior work is methodological: earlier studies did not consistently compare against the correct base model, and did not tune prompts per model, so an apparent domain gain could be a prompt-fit artefact.

This is the single most important paper in this review for how we run experiments. Three consequences we should adopt as standing practice:

1. **Always evaluate the base model under the identical harness.** A CPT result is a *delta*, and a delta needs a control run, not a published number.
2. **Report the scoring convention.** Our evaluator supports letter-scoring and text-scoring precisely because base models are badly served by letter-scoring; reporting one without the other invites the artefact Jeong et al. describe.
3. **Report uncertainty.** MedQA's test split is 1,273 items; a 1.5-point difference there is well inside noise. Confidence intervals are not optional.

Read charitably, this paper does not say domain adaptation cannot work. It says the field has not yet demonstrated that it does, under fair comparison. That is a *better* motivation for this project than the optimistic framing: if generic medical CPT gives little, then showing that *selected* medical CPT gives more is a genuinely open and worthwhile result.

### 3.6 The 2026 evidence: still contested, in both directions

A deep research pass in July 2026 (25 claims extracted from 2025-2026 sources, each put through a 3-vote adversarial verification, 16 survived) found the contrarian result neither settled nor overturned. Both directions strengthened.

**Extending the contrarian finding.** LiveMedBench [Yn26], a contamination-free suite refreshed weekly from real clinical cases, reports MedGemma 27B scoring 5.9% against 6.4% for the general-purpose Gemini 2.5 Flash, a direct 2026 extension of Jeong et al.'s result to a newer medical model on a newer benchmark. Dorfner et al., peer-reviewed in JAMIA [Do25] (arXiv v1 predates this window but the peer-reviewed version is 2025), independently find biomedical fine-tuning adds little on unseen clinical data under a fair comparison. MedCheck [Ch26] audited 56 medical LLM benchmarks against 46 lifecycle criteria and found 49 of 56 do not address contamination at all, and names MedQA and MedMCQA explicitly as paradigm cases of convenience-driven benchmark design, exam questions standing in for clinical data.

**Complicating it.** Medmarks [Wa26], a 30-benchmark open suite, reports medically adapted models beating their own bases with mean win-rate gains of 0.32 to 0.55 depending on model family, disconfirming evidence against the contrarian framing on this particular suite. Domingo-Aldama et al. [Dm26] find medical adaptation pays off specifically for under-served languages, a nuance rather than a reversal. Read together with Jeong et al., the honest summary is: fair comparison still finds medical adaptation an inconsistent win, model-family and benchmark dependent, not a settled loss. The project's reframed question, does *selected* CPT help where unselected CPT does not, remains open and is not weakened by either direction.

**A benchmark-validity complication that cuts across both directions.** MedCheck [Ch26] also reports that 34% of the 56 audited benchmarks evaluate only a single dimension such as accuracy, and 50% align with no formal medical standard such as ICD or SNOMED CT. Any single number from MedQA, MedMCQA or PubMedQA, in either direction, should be read against that backdrop.

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## 4. Stage 2: Supervised fine-tuning

### 4.1 What SFT does

SFT trains on instruction-response pairs with the loss masked to the response span, so the model is not rewarded for reproducing the prompt. Its role is format and instruction-following, not primarily knowledge acquisition.

For our three QA datasets, SFT is where they naturally live: MedQA and MedMCQA supply question-with-options → answer, and PubMedQA supplies question-with-abstract → yes/no/maybe plus a long-form rationale. MedMCQA additionally ships explanations (exp) for a subset of rows, which makes chain-of-thought-style SFT possible without generating rationales synthetically.

## 4.2 The combined-objective question

A design question raised in this project's own planning, and worth treating as a research question rather than an implementation detail: should the instruction stage be trained with a **combined objective** (next-token loss on raw biomedical text *alongside* instruction loss on QA pairs) instead of running CPT and SFT strictly in sequence?

The motivating intuition is that sequential SFT erodes the domain knowledge CPT just installed, and that mixing keeps it. This is essentially the replay idea from Section 3.2 applied across the CPT/SFT boundary rather than within CPT. There is supporting work on balancing continued pretraining against instruction tuning, and the general replay evidence is strong, but we should treat the specific medical claim as untested and design an ablation for it. It is recorded in the codebase at `src/medsel/stages/sft.py` so it does not get lost.

## 4.3 SFT data selection

This is where most of the recent selection literature actually operates (Section 7). LESS [X24a] and 3DS [D25] are both SFT-stage methods. Their headline results are strong: LESS reports that 5% of the data, selected well, can beat the full dataset, which is the strongest available evidence that selection is worth doing at all.

UltraMedical [Zg24] is worth naming as context rather than as a method: a large curated biomedical instruction corpus (NeurIPS 2024 Datasets and Benchmarks Track, Spotlight) built by synthesis and preference filtering rather than selection from a fixed pool. It shows the field's SFT-stage answer to data scarcity is generation, not only selection, an alternative worth weighing against selecting from the three QA sets already in scope.

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## 5. Stage 3: Preference optimisation

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Covered briefly, because it is the furthest away and because we have a data problem before we have a method problem.

**Methods.** DPO is the practical choice at our scale: it optimises directly against preference pairs and needs no separate reward model or online sampling loop, unlike PPO-style RLHF. TRL provides a `DPOTrainer` that works against the same model and tokenizer objects the earlier stages use.

**The blocker is data, not code.** The proposal says preference data "if available", and for the public datasets in scope, it is not. Options, in rough order of cost:

1. Construct pairs from PubMedQA: the expert `long_answer` as chosen, a model-generated or perturbed answer as rejected. Cheap, but the "rejected" side is artificial and the resulting preference signal may be about fluency rather than clinical correctness.
2. Use MedMCQA explanations similarly, with the correct-option explanation as chosen.
3. Adopt an existing open medical preference set, if one of adequate quality exists. The survey below answers this: no medical-specific set was found, but general open preference corpora are concrete and usable now.

**The survey.** Djuhera et al. [Dj26] audit five open, non-credentialed DPO corpora: TuluDPO (around 280k pairs), ORPO (`mlabonne/orpo-dpo-mix-40k`, 40k pairs), UltraFeedback, HelpSteer, and

Code-Preference-Pairs. Two findings matter directly for option 3:

- **Curated beats largest.** UltraMix, a 190k-pair mixture assembled selectively across all five sources, is 30% smaller than the strongest individual source (TuluDPO) yet scores higher across 14 evaluation tasks. This is itself a preference-stage selection result: curating beats using the single best full dataset, which is the same shape of claim the proposal makes for CPT and SFT.
- **These corpora are noisy at the pair level.** Only 70 to 80% of pairs across TuluDPO, ORPO, UltraFeedback and HelpSteer have a reward-model preference ordering that agrees with the labelled chosen answer, meaning roughly 20 to 30% of pairs may carry a questionable or inverted label. Code-Preference-Pairs is the exception, with better label agreement. Any DPO run built on these sources inherits this noise floor and should not treat the labelled preference as ground truth without a sanity check.

None of the five is medical, so option 3 does not remove the need for options 1 and 2; it gives a general-domain preference set that could be mixed with medically-constructed pairs, the same replay logic as Section 3.2 applied to Stage 3.

Safety and hallucination are the capabilities this stage is normally credited with, and they are also the ones our current benchmarks (three MCQ datasets) cannot measure at all. Section 6.5 lists concrete open benchmarks that do measure some of this. That gap should be stated plainly in any write-up.

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## 6. The datasets

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Full field-level schemas, split semantics, and the traps we hit are in [datasets.md](#). This section covers provenance and what each dataset is good for.

### 6.1 MedQA (owner: *Debmalya*)

USMLE-style clinical vignettes with four options [J21]. Questions are long (a full patient presentation) and require multi-step clinical reasoning rather than fact recall, which makes it the hardest of the three for small models.

Widely reported figures: GPT-4 around 86% and Medprompt-style prompting above 90%; the strongest open models via prompt engineering around 72.6% (Yi-34B, OpenMedLM) (*secondary*). A 1B base model should be expected near chance (25%) before any adaptation, which is worth stating so that a smoke-test number is not mistaken for a failure.

Caveats: the Hub mirror we use ships an empty `subject_name` column, so MedQA offers **no per-subject breakdown**. It cannot contribute to per-capability analysis, only to aggregate accuracy.

### 6.2 MedMCQA (owner: *Animesh*)

~194k questions from Indian medical entrance exams (AIIMS, NEET-PG), spanning 21 subjects and ~2.4k topics [P22]. Its size makes it the only one of the three usable as a substantial SFT source, and its `subject_name` / `topic_name` labels make it **the only per-capability signal any of our datasets ships**. That is why the evaluator's subject breakdown keys on it.

The important structural fact: the published **test split withholds gold answers** (`cop = -1`). Every comparable number in the literature is measured on `validation` (the NEET-PG set), and so is ours. Reported open-model SOTA is around 68.3% (*secondary*).

Quality caveat worth documenting: MedMCQA is known to contain some noisy items (typos, ambiguous phrasing, and occasional disputed keys), a consequence of its scraped origin. This is

relevant to us twice over: it is an argument *for* data selection at the SFT stage, and it is a reason not to over-interpret small accuracy differences.

### 6.3 PubMedQA (*owner: Arkajyoti*)

Research questions derived from PubMed article titles, answerable yes/no/maybe from the abstract [J19]. Three configs:

- `pqa_labeled`: 1,000 expert-annotated items, the evaluation set
- `pqa_artificial`: ~211k auto-labelled items, usable for SFT
- `pqa_unlabeled`: ~61k with long answers but **no final\_decision column at all**

Two things make PubMedQA different from the other two: it is a 3-way decision rather than a 4-way MCQ, and it supplies *context* (the abstract), so it measures reading comprehension over a given passage rather than parametric knowledge.

**Comparability caveat.** The canonical 500-question test set is defined by a ground-truth file in the authors' repository, which the Hub mirror does not carry. `pqa_labeled` arrives as one undivided 1,000-row split. Our numbers are therefore internally consistent but should not be quoted against published PubMedQA figures without adopting the official split.

**Contamination caveat.** PubMedQA is built from PubMed abstracts, and our CPT corpus is PubMed abstracts. Any CPT-then-evaluate-on-PubMedQA experiment must exclude the PubMedQA PMIDs from the CPT corpus, or the result is meaningless. This is a concrete task, not a caveat to note and forget.

### 6.4 The PubMed CPT corpus (*owner: Srinjoy*)

We use MedRAG/pubmed [X24b], the corpus assembled for the MedRAG benchmark: 23.9M PubMed articles that have both a title and an abstract, pre-chunked into 1,166 JSONL shards.

Measured properties (verified directly against the Hub, not quoted):

- 1,166 shards, **70 GB total**, averaging ~60 MB per shard
- ~296 tokens per snippet on average
- fields: `id`, `title`, `content` (abstract), `contents` (= `title` + " " + `content`), `PMID`

Choosing this over the NCBI baseline FTP trades some control for a lot of avoided work: the XML is already parsed, filtered to articles with usable abstracts, and chunked. The costs of that choice, which we should state in any write-up: it is a snapshot (the `pubmed23` naming indicates a 2023 baseline), it is abstracts-only rather than full text, and its inclusion filter is MedRAG's rather than ours.

If full text becomes necessary (plausible for long-context or reasoning-oriented CPT), PubMed Central Open Access is the successor corpus, and MEDITRON's corpus construction [C23] is the reference for how to do it.

### 6.5 Beyond MCQ: benchmarks for the other capabilities

Open Question 5 (Section 10) names the project's admitted weakness: the proposal claims six clinical capabilities, and MedQA, MedMCQA and PubMedQA measure roughly one and a half. A 2026 research pass looked specifically for open, non-credentialed benchmarks covering the rest, since this project cannot use MIMIC-IV-style credentialed data. Four are concrete and usable now:

| Benchmark          | Capability covered                                                        | Access                                                                                                                            | Headroom in 2026                                                                                                                         |
|--------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| HealthBench [Op25] | Open-ended clinical response quality, safety, communication               | CC BY-4.0, <a href="https://github.com/openai/simple-evals">github.com/openai/simple-evals</a>                                    | Main split moved 16% to 60% in two years; HealthBench Hard still sits at 32% for the best model at release, so it has not saturated      |
| BRIDGE [Wu25]      | Information extraction, ICD-10 coding, summarisation, NLI, classification | Open, 87 tasks over 59 real clinical text datasets                                                                                | ICD-10 coding accuracy around 15%, generation tasks around 20%, large headroom                                                           |
| ACI-Bench [Yi23]   | Clinical summarisation (visit-note generation from dialogue)              | CC BY 4.0, no credentialing, Figshare DOI <a href="https://doi.org/10.6084/m9.figshare.22494601">10.6084/m9.figshare.22494601</a> | Only 207 dialogue-note pairs total (40 per test split), too small to power a regression benchmark; usable as a qualitative probe only    |
| Medmarks [Wa26]    | QA, information extraction, medical calculation, open-ended reasoning     | Open, 30 benchmarks, includes an RL-trainable split (Medmarks-T)                                                                  | Reports the MultiMedQA suite (MedQA, MedMCQA, PubMedQA's core) as mostly saturated, which independently confirms why this section exists |

Two more are worth knowing about without adopting outright: LiveMedBench [Yn26] refreshes weekly from real clinical cases specifically to stay contamination-free, at the cost of being a moving target rather than a fixed benchmark a thesis can cite a stable number against; MedXpertQA [Zu25] is a harder MCQ-style benchmark (expert-level reasoning) that would raise the difficulty ceiling of the existing MCQ evaluation without adding a new capability.

The practical implication: capability coverage no longer needs a custom benchmark built from scratch to grow past MCQ. The gap is integration effort (a fourth loader shape, since none of these four match `QAExample`), not the absence of a benchmark. This should be weighed against the month 9-10 validation timeline named in Section 10.

## 7. Data selection methods

The organising question: given a large pool of candidate training data and a budget, which subset should we train on? The literature splits by *what signal* is used to answer it.

### 7.1 Influence-based

The idea: estimate how much each training example changes performance on a target set, and keep the most helpful.

- **Influence functions** [K17]. The foundational work, approximating the effect of upweighting a training point via the inverse Hessian. Exact computation is intractable at LLM scale; everything since is an approximation to this.
- **TRAK** [P23]. Makes influence attribution tractable at scale by linearising the model and using random projections, with ensembling to reduce variance. Practical for attributing model behaviour, but still expensive.
- **LESS** [X24a]. The most directly relevant to us. It adapts influence formulations to work with **Adam** (not plain SGD) and with variable-length instruction data, builds a reusable low-rank **gradient datastore**, and then selects examples by gradient similarity to a handful of few-shot examples that embody a target capability. Two results matter for this project:
  - training on a LESS-selected **5%** can outperform training on the full dataset;



- selections **transfer**: a small model can pick data for a larger one, and across families.

That transfer property is what makes influence-based selection affordable for a student project: we can score with a 1B model and train something bigger.

## 7.2 Optimisation-based subset selection

- **GLISTER** [K21a]. Frames subset selection as bilevel optimisation against a held-out validation set, with a greedy approximation.
- **GradMatch** [K21b]. Selects a subset whose *summed gradient* approximates the full-data gradient, via orthogonal matching pursuit.

Both predate LLM-scale instruction tuning and were demonstrated on smaller models, but the gradient-matching objective is conceptually clean and is a reasonable thing to revisit at our scale.

## 7.3 Distribution matching and heuristic quality

- **DSIR**: importance resampling to match a target distribution in a hashed n-gram feature space. Very cheap, and the natural baseline for corpus-scale (CPT) selection where per-example gradients are unaffordable.
- **DataComp** [G23] and **DataComp-LM** [L24]. Rather than proposing a method, these fix the model and training budget and make *the dataset* the submission. The lesson we should copy is methodological: a selection result is only meaningful against a fixed training budget and a fixed evaluation suite. Our `cpt_baseline.yaml` exists for this reason.
- **Quality classifiers**. The FineWeb-Edu / phi lineage trains a lightweight classifier on model-labelled "educational value" and filters the corpus with it. Cheap at corpus scale and a strong baseline for CPT selection.
- **Deduplication**. Exact and near-duplicate removal is consistently among the highest value-per-compute interventions in pretraining-data work. For PubMed specifically, duplicate deposits, errata, and near-identical abstracts are real.

## 7.4 Medical-domain selection

**3DS** [D25] is the closest published work to our project and should be treated as the primary baseline to beat. It is a two-stage method for medical domain adaptation:

1. **Prompt-driven data selection** filters noise by checking alignment with what the model already knows.
2. **Decomposed difficulty-based selection** scores each example on three metrics (*instruction understanding*, *response confidence*, and *response correctness*) with attention-based importance weighting for calibration.

Reported to beat prior methods by up to 2.97% accuracy in healthcare, with validation in law and general domains (*secondary*). Code and data are open.

Note what 3DS is and is not: it is **SFT-stage**, **single-stage**, and optimises **accuracy**. It decomposes *difficulty* into three axes, which is genuinely multi-signal, but it does not decompose the *objective* into multiple capabilities, and it says nothing about CPT. That is exactly the gap the proposal targets.

## 7.5 Summary table

| Method                    | Stage                  | Signal                                                | Cost                                         | Multi-target?                                           |
|---------------------------|------------------------|-------------------------------------------------------|----------------------------------------------|---------------------------------------------------------|
| Influence functions [K17] | any                    | Hessian-approx influence                              | intractable at scale                         | no                                                      |
| TRAK [P23]                | any                    | linearised attribution                                | high                                         | no                                                      |
| LESS [X24a]               | SFT                    | Adam-aware gradient similarity                        | moderate, amortised datastore                | no (single target set)                                  |
| GLISTER [K21a]            | SFT                    | bilevel validation loss                               | moderate                                     | no                                                      |
| GradMatch [K21b]          | SFT                    | gradient matching                                     | moderate                                     | no                                                      |
| DSIR                      | pretrain               | n-gram distribution match                             | very low                                     | no                                                      |
| Quality classifier        | pretrain               | learned quality score                                 | low                                          | no                                                      |
| 3DS [D25]                 | SFT (medical)          | decomposed difficulty                                 | moderate                                     | partially (3 difficulty axes)                           |
| BETR [Be25]               | pretrain               | benchmark-embedding similarity, distilled to fastText | low at inference, one-time distillation cost | partially (diverse-target variant, one aggregate score) |
| Data Mixing Agent [Ya26]  | CPT                    | learned domain-mixture policy (offline RL)            | moderate, one-time policy training           | partially (multi-domain, single stage)                  |
| Meta-rater [Zh25]         | pretrain               | multi-dimensional learned quality score               | low at inference                             | no                                                      |
| <b>This project</b>       | <b>CPT + SFT + DPO</b> | <b>multiple, stage-specific</b>                       | <b>TBD</b>                                   | <b>yes, by construction</b>                             |

## 7.6 Corpus-scale selection, 2026 update

Four results from a 2026 research pass bear directly on the CPT stage, and one of them changes how `resolve_budget` should be reasoned about.

**The optimal selection ratio scales with compute, not a constant.** BETR [Be25] fits  $F_{\text{opt}}(C) = 4e-5 \times C^{0.25}$ , the fraction of the corpus worth keeping, rising from about 3% at  $1e20$  FLOPs to about 30% at  $1e23$  FLOPs. Larger training runs need *less* aggressive filtering. This is falsifiable and directly load-bearing: it means a keep-rate copied from a large published recipe is provably wrong at this project's scale, and a small-compute CPT run should filter far more aggressively than DataComp-LM or similar large-scale work would suggest.

**Selecting for one target measurably costs the others.** BETR's own ablation is the clearest empirical support in this review for the multi-target premise: a variant targeted at one benchmark suite (Core) is best on that suite but falls to third place on held-out (Noncore) tasks. This confirms Gap 2 is a real cost, not a hypothetical one, though it also means BETR itself already occupies part of the multi-target space (Section 8 below).

**A domain-specific signal beats a generic quality classifier for medicine.** Huang et al. [Hu26], at matched compute on a French medical encoder, find a medical-term density filter beats the standard educational-quality classifier as a single-axis filter (78.27 versus 59.16 win probability against an unfiltered baseline of 45.02), and the intersection of both beats either alone. This is the first published, concrete result naming a domain-specific corpus-scale filter for medical text, and it is a stronger baseline to beat than a generic FineWeb-Edu-style classifier for this

project's CPT stage. Caveat carried with the claim: encoder MLM in French, not decoder CPT in English, and compared against unfiltered data, not a random subsample of equal size.

**The scariest-looking result does not license skipping selection here.** Mohri et al. [Mo26] report unfiltered web data beating every quality filter tested, at model and data scale. The crossing point where unfiltered data wins is real, but the authors' own projection puts it near  $1e30$  FLOPs for the full 240T-token pool, against roughly  $5e26$  FLOPs for a current frontier pretraining run, and they state plainly that "when compute is a bottleneck, we expect filtering to still be important." Below a model-size threshold the crossing does not occur at all. This project's compute budget sits nowhere near the regime where the result applies.

**A filtered corpus can beat the clean corpus it was filtered toward.** Nait Saada et al. [Na26] report a classifier-filtered web corpus scoring 53.8% against 50.1% for a model trained directly on the unlimited high-quality reference set under identical conditions (350M parameters). The useful reading for us is narrow: having a small clean in-domain set does not mean training on it is the best use of it, and using it as a filter over a large pool can do better.

The same paper's account of *why* filtering helps, that the benefit is noise removal rather than resemblance to the target distribution, did not survive adversarial verification in our research pass and is deliberately not relied on here. Treat the mechanism as open and the 53.8 versus 50.1 comparison as the citable result.

## 7.7 Preference-stage data selection

Section 5 covers this in detail as the answer to the Stage 3 data blocker. In selection terms, the one-line summary from Djuhera et al. [Dj26] is that curated preference data beats the largest single source: a 190k-pair mixture assembled across five open corpora beats the strongest 280k-pair single source on 14 evaluation tasks, the same "selection beats scale" claim this project makes for CPT and SFT, now demonstrated for DPO too, on general-domain rather than medical data.

For a broader map of data-centric efficient training methods than this section covers, Luo et al.'s 2025 survey [Lu25] is worth citing wholesale rather than re-deriving.

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## 8. Gap analysis

Reading across Sections 3-7, four gaps were consistent as of the original review. A 2026 research pass revisited each adversarially, specifically to check whether anyone had closed them since. Two were narrowed rather than closed; two stand as before; and one new gap emerged from the same pass.

**Gap 1: Single-stage scope. Narrowed, not closed.** Nearly all selection work targets SFT. LESS, GLISTER, GradMatch and 3DS are all instruction-tuning methods. CPT-stage selection is dominated by cheap heuristics (dedup, quality classifiers, distribution matching) because per-example gradient methods do not scale to 23.9M documents. The live threat is He et al. [He26], published June 2026, which states the same load-bearing premise this project makes, that different training stages want different data because they are credited with different learning roles, and instantiates it as a difficulty split between SFT and RL data. It has no CPT stage, single capability (math and logic reasoning), and two small non-medical models. Meta-rater [Zh25], an ACL 2025 Best Theme Paper on multi-dimensional pretraining selection, and Holistic Data Scheduler [Da26], KDD 2026, both remain scoped to a single stage and never ask whether the best pretraining subset depends on downstream SFT data. **The honest claim after this pass: someone now owns the SFT-to-RL half of Gap 1. Nobody has published a policy spanning CPT through SFT through DPO, which is what this project claims.**

**Gap 2: Single-objective optimisation. Under active contest, needs sharpening.** Selection is almost universally optimised against aggregate downstream accuracy on one benchmark. Three 2026 papers now occupy part of this space and must be distinguished from, not just cited past: Data Mixing Agent [Ya26] explicitly balances performance across multiple domains (general versus math, general versus code) during CPT and reports cross-domain variance as a metric, all tested reweighting methods cut variance by over 200 versus the base model. Holistic Data Scheduler [Da26] ships under the literal phrase "multi-objective data selection," though its three objectives are internal training signals (a quality score, inter-domain loss influence, weight norms), not downstream capabilities. BETR [Be25] has a diverse-target variant that matches or exceeds single-target baselines on disjoint benchmarks. **None preserves several NAMED clinical capabilities separately; all collapse their targets into one aggregate score or a small number of coarse domains, and none combine multi-objective scope with the multi-stage scope of Gap 1.** The project can no longer claim multi-target selection is unexplored at pretraining scale. It can still claim that preserving named, distinct capabilities across stages is unexplored, which is a narrower and more defensible claim than the original phrasing, and a reviewer familiar with Data Mixing Agent or HDS will expect this distinction to be made explicitly.

**Gap 3: Long-tail and rare-event coverage is not a constraint. Unchanged.** Selection methods rank and truncate. Anything scoring low is dropped, and rare diseases and underrepresented populations score low almost by definition, because they are rare. Making coverage a *constraint* rather than something the ranking might incidentally preserve is a genuine methodological difference, and it is why the harness separates scoring from selection and gives `select_stratified` a `min_per_group` floor. Nothing in the 2026 pass addressed long-tail coverage as a constraint rather than an incidental ranking outcome.

**Gap 4: The baseline may be weaker than reported. Unchanged, evidence on both sides grew.** Per Jeong et al. [J24], the premise that medical CPT reliably helps is not well established. Section 3.6 covers the 2026 follow-up in full: LiveMedBench [Yn26] and Dorfner et al. [Do25] extend the contrarian finding to newer models and a fair clinical-task comparison, while Medmarks [Wa26] and Domingo-Aldama et al. [Dm26] complicate it without reversing it. Rather than undermining the project, this still sharpens it: the interesting claim is not "CPT helps" but "*selected* CPT helps where *unselected* CPT does not", and that comparison remains one nobody has run.

**Gap 5: Benchmark and evaluation validity, new in this pass.** MedCheck [Ch26] audited 56 medical LLM benchmarks against 46 lifecycle criteria and found 49 of 56 (88%) do not address data contamination at all, 34% evaluate only a single dimension such as accuracy, and MedQA and MedMCQA are named explicitly as paradigm cases of exam questions substituting for real clinical data. This is not a new instance of Gap 4, it is a claim about the measuring instrument rather than the model: even a correctly-run base-model control, scored on MedQA and MedMCQA, is being scored against benchmarks whose own field-wide audit found wanting on contamination handling and dimensionality. Section 6.5 names four open, non-credentialed benchmarks that partially address this by measuring capabilities beyond MCQ; adopting even one would directly answer part of this gap. CapTrack [Th26] supplies a reusable framework for measuring what MedCheck's audit found underserved: it decomposes retention into three groups (latent competence, behavioural preference, protocol compliance) rather than a single aggregate score, which is closer to the per-capability measurement this project needs than accuracy alone.

A sixth, more practical gap: **auditability**. The proposal promises interpretable data-valuation scores and explanations for why a sample was kept or discarded. Almost nothing in Section 7 produces a human-readable justification; influence scores are numbers without narratives. In a clinical setting that is a real deficiency and a defensible contribution on its own. Fully Open Meditron [Te26] is a concrete 2026 precedent worth reading for how an auditable clinical LLM

pipeline can be built and documented, though it targets pipeline transparency rather than per-sample selection justification, so it does not close this gap either.

## 9. Implications for our harness

Design decisions in this repository that follow directly from the review:

| Decision                                                                                         | Source                                                                                                               |
|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| CPT targets <b>Gemma 3 base</b> , not MedGemma                                                   | §3.3: adapting an already-adapted model makes gains unattributable                                                   |
| An <b>unselected CPT baseline</b> config ships alongside any selection experiment                | §7.3 DataComp; §3.5 Jeong                                                                                            |
| Evaluator supports <b>letter and text scoring</b> , reports both acc and acc_norm                | §3.5: prompt/scoring artefacts                                                                                       |
| Evaluator reports <b>per-subject accuracy</b>                                                    | §8 Gap 2; MedMCQA is the only per-capability signal (§6.2)                                                           |
| Scoring and selection are <b>separate modules</b>                                                | §8 Gap 3: coverage is a selection constraint, not a score                                                            |
| <code>select_stratified(min_per_group=...)</code>                                                | §8 Gap 3: rare groups must survive aggressive budgets                                                                |
| Baseline scorers are <b>random and length</b>                                                    | §7: a method that cannot beat random has shown nothing                                                               |
| Prompt templates are <b>shared and versioned</b>                                                 | §3.4: otherwise data effects and format effects confound                                                             |
| <b>PMID-level exclusion</b> required before PubMedQA evaluation                                  | §6.3 contamination                                                                                                   |
| <code>dedup</code> on by default in the PubMed loader                                            | §7.3: dedup is high value per unit compute                                                                           |
| CPT selection ratio should scale with compute rather than being a fixed constant                 | §7.6: $F_{\text{opt}}(C) = 4e-5 \times C^{0.25}$ [Be25]                                                              |
| A domain-term-density scorer is a stronger CPT baseline than a generic quality classifier        | §7.6: medical-term density beat educational quality on a matched-compute medical corpus [Hu26]                       |
| Gap 1 and Gap 2 claims must be stated as the narrower, cross-stage and named-capability versions | §8: both gaps were partially occupied by 2026 work [He26] [Ya26] [Da26]                                              |
| A non-MCQ benchmark should be adopted before month 9-10 validation, not deferred                 | §6.5, §8 Gap 5: field-wide audit found MCQ-only evaluation and single-dimension scoring a documented weakness [Ch26] |

Concrete near-term work items this review generates, beyond the proposal's plan:

1. **Build the PMID exclusion list** from PubMedQA `pqa_labeled` (and `pqa_artificial` if used for SFT) and wire it into the PubMed loader as a filter. Until this exists, no PubMedQA number from a CPT'd model is trustworthy.
2. **Run the base-model control** for every benchmark before any CPT run, and store it as a committed reference result.
3. **Add bootstrap confidence intervals** to `EvalReport`. MedQA's 1,273-item test set cannot support the precision people routinely claim on it.
4. **Adopt at least one non-MCQ benchmark** from Section 6.5 (HealthBench, BRIDGE, ACI-Bench or Medmarks) rather than deferring capability coverage past validation. The survey this

item used to depend on is now done.

5. **Build a compute-aware selection budget** rather than a fixed keep-rate, following the  $F_{\text{opt}}(C)$  relationship in Section 7.6, and treat medical-term density as the baseline the CPT scorer must beat rather than a generic quality classifier.
  6. **Use the Djuhera et al. [Dj26] preference corpora as a general-domain replay source for Stage 3**, mixed with the medically-constructed pairs from options 1 and 2 in Section 5, rather than waiting for a medical-specific preference set that a 2026 survey did not find.
- 

## 10. Open questions

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1. Can gradient-based influence (LESS-style) be made affordable at CPT scale, or must CPT selection stay in the cheap-heuristic regime? The transfer result in [X24a], where small models pick data for large ones, is the most promising lead.
2. Is CPT selection *conditionally* optimal? That is, does the best CPT subset change depending on the SFT data that follows? If yes, that alone justifies the multi-stage framing.
3. Does the combined CPT+instruction objective (§4.2) actually retain domain knowledge better than sequential CPT→SFT?
4. What is the right unit of selection for CPT: the document, the packed block, or a topic cluster? Our packing implementation currently makes the document the unit.
5. How should capability coverage be *measured* when we only have MedMCQA subject labels and PubMedQA MeSH terms to work with? Everything else in the proposal's capability list (diagnosis, summarisation, ICD coding, temporal reasoning) has no benchmark in our current set. **Update, 2026:** this is now partly answerable rather than fully open. Section 6.5 names four open, non-credentialed benchmarks (HealthBench, BRIDGE, ACI-Bench, Medmarks) that between them cover information extraction, ICD-10 coding, summarisation and open-ended clinical reasoning. The question narrows from "does a usable benchmark exist" to "which one to integrate first, and whether integrating one is enough to defend the six-capability claim."
6. Given that He et al. [He26] and Data Mixing Agent [Ya26] each occupy part of Gap 1 or Gap 2 individually, does *combining* stage-awareness and multi-capability preservation in one policy, the project's actual claim, produce a result neither paper's narrower version would predict? If the combination is separable into independent per-stage and per-capability effects, the project's contribution is the engineering of the combination rather than a new empirical finding, and that distinction should be decided before, not after, running experiments.
7. Baek et al. [Ba26] report that moving the small domain dataset conventionally reserved for finetuning into the pretraining mixture ("specialized pretraining") cuts the pretraining tokens needed to reach a given domain performance by up to 1.75x, on ChemPile, MusicPile and ProofPile (*secondary: figure taken from the abstract, and none of those domains is medicine*). Does the same hold when the domain shift is general web text to biomedical abstracts? If it does, the question it raises for this project is whether stage boundaries are the right unit of decomposition at all, or whether the better question is where in one continuous run each kind of data belongs. Note that a stronger reading of this paper, that it argues normatively for blurring the stage boundary, did **not** survive adversarial verification, so the challenge here is one we are inferring from the result rather than one the authors make.

Question 5 was the most uncomfortable one in the original review, and is now the one where a 2026 research pass changed the answer most: the proposal names six clinical capabilities; the three datasets measure roughly one and a half of them, but usable open benchmarks for several of the rest now exist and do not require new construction. Question 6 is the sharper novelty question the project must now answer given Section 8, since two separate 2026 papers each independently occupy half of the claimed novelty. Question 7 is the more fundamental challenge, since it

questions the stage-separation premise directly rather than the coverage of existing selection methods. All three should be resolved before the month 9-10 validation phase, not during it.

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## 11. References

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Keyed to `references.bib`. Entries tagged 2026-07 came from a deep research pass that post-dates the original review; see Section 3.6, 5, 6.5, 7.6, 7.7 and 8 for how each is used.

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- **[Hu26]** Huang et al. *Where Does the Signal Live? A Web Data Recipe for Medical Encoder Pretraining*. 2026. arXiv:2606.22079. (2026-07)
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- **[L24]** Li et al. *DataComp-LM: In Search of the Next Generation of Training Sets for Language Models*. 2024. arXiv:2406.11794.
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# Appendix: Deep Research Findings, July 2026

This appendix records the raw output of the research pass that produced the 2026 additions to the review above. It is included so the evidence can be checked rather than taken on trust.

**Method.** Five parallel search agents covered corpus-scale selection, the multi-stage and multi-target novelty check, follow-ups to the contested premise that medical adaptation helps, capability retention and non-MCQ benchmarks, and an open sweep. Twenty-five sources were read and 125 candidate claims extracted. Each claim that reached verification was then given to three independent agents instructed to *refute* it, with a majority refutation killing the claim. Sixteen of twenty-five survived.

**How to read this.** Surviving claims are evidence, not proof: they survived an attempt to break them. Killed claims are listed too, because a reader who finds the same paper independently should know the claim was tested and did not hold. Where a figure came from an abstract rather than a paper body, the claim says so.

## Surviving claims (16)

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**S1.** The no-filter-wins result is gated behind a compute threshold that no realistic academic project can reach: two independently fitted compute scaling laws (R-squared above 0.99) predict that unfiltered 240T-token DCLM-Pool only overtakes RefinedWeb at roughly  $1e+30$  FLOPs, versus a current frontier pretraining estimate of about  $5e+26$  FLOPs. The authors explicitly concede filtering remains important when compute is the bottleneck, which means this paper does NOT license abandoning selection on a small or single-GPU budget.

Source: <https://arxiv.org/abs/2605.19407> (primary) · refuted by 0 of 3 verifiers

*The compute required for raw Common Crawl to outperform our tested filters is large, up to around  $1e30$  FLOPs with our projections in Figure 6. When compute is a bottleneck, we expect filtering to still be important.*

---

**S2.** At matched compute (20B pretraining tokens per configuration, French medical encoder MLM), a domain-targeted medical-term density filter beats the standard educational-quality classifier as a single-axis filter, and intersecting the two beats both single filters and both curated medical corpora. Body numbers (Table 1, Min-Max / Win Probability aggregates over 7 DrBenchmark tasks): unfiltered FineWeb-2-Medical 45.02 / 45.45; educational quality  $\geq 4$  gives 59.16 / 40.26; medical-term density  $\geq 0.1$  gives 78.27 / 66.23; the intersection Edu  $\geq 4$  AND Med-term  $\geq 0.1$  gives 81.25 / 68.83, described in the text as beating external baselines by +29 WP points. Direct evidence that a cheap corpus-scale quality filter beats a strong unfiltered baseline, and that a generic quality classifier is the wrong signal for a dense-terminology domain. Caveat: encoder MLM in French, not decoder CPT in English, and the comparison is to an unfiltered baseline, not to a random subsample of equal size.

Source: <https://arxiv.org/abs/2606.22079> (primary) · refuted by 0 of 3 verifiers

*Among single-axis filters, medical-term density is strongest, subdomain second, educational quality weakest. For decoder LLMs, educational quality is a dominant filter signal; for encoder MLM in a dense-terminology domain, signals targeting domain-token concentration carry more weight. [...] The intersection of educational quality and medical-term density is the strongest filter: it outperforms medical-term density alone and beats external baselines by +29 WP points.*

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**S3.** Selecting data against one target benchmark suite measurably degrades performance on held-out tasks: BETR Target-Core is best on the Core suite it targeted but drops to third place on the held-out Noncore tasks. This is the single most useful result in the paper for the project's multi-target premise, and it cuts both ways. It empirically confirms that a selection policy optimised for one capability target trades away others (supporting Gap 2, single-objective optimisation, being a real problem), but it also means BETR already partially occupies the multi-target space, since the paper explicitly evaluates a diverse-target variant that 'matches or exceeds baselines' on disjoint benchmarks. The project cannot claim multi-target selection is unexplored at pretraining scale; it must claim the finer point that BETR still collapses its targets into one aggregate similarity score rather than preserving named capabilities separately.

Source: <https://arxiv.org/abs/2507.12466> (primary) · refuted by 0 of 3 verifiers

*BETR Target-Core achieves the best Core performance but falls to third place on held-out Noncore tasks.*

---

**S4.** A learned, model-based domain reweighting policy (Data Mixing Agent, trained with offline RL / Conservative Q-Learning on sampled data mixing trajectories) beats the static regression-based mixing baseline RegMix by 3.02% average across 12 benchmarks (8 general + 4 math) during continual pretraining of LLaMA-3B-DCLM-100B, and beats the base model by 8.88%. Verified in the paper body (Section 4.2 and Appendix), not only the abstract. This is direct 2026 evidence that corpus-scale mixture optimisation for CPT still yields measurable gains over strong non-random baselines, and it post-dates every mixing citation currently in the review.

Source: <https://arxiv.org/abs/2507.15640> (primary) · refuted by 0 of 3 verifiers

*Overall, DataAgentRL achieves 47.03% on average, surpassing RegMix by 3.02%, DBL by 3.53%, and the base model by 8.88%. DataAgentRL also outperforms DataAgentSFT by a large margin of 2.08%.*

---

**S5.** DISCONFIRMING for the multi-target novelty claim (Gap 2): this ACL 2026 paper explicitly optimises balanced performance across multiple fields/capabilities simultaneously rather than aggregate accuracy on one benchmark, and reports variance across fields as an evaluation metric (all reweighting methods cut performance variance by over 200 versus the base model; the agent is best on variance). The niche is only partially occupied though, because the multiple objectives are coarse DOMAINS (general vs math, general vs code) inside a SINGLE stage (CPT only), with no SFT or DPO stage and no clinical capability decomposition. Gap 2 is narrowed, not closed; Gap 1 (stage-awareness across CPT/SFT/DPO) is untouched by this work.

Source: <https://arxiv.org/abs/2507.15640> (primary) · refuted by 0 of 3 verifiers

*All domain reweighting methods significantly reduce performance variance by over 200 compared to the base model. Data mixing agent further achieves the best results in variance, further demonstrating its advantage in stable performance across fields. ... Overall, DataAgentRL can coordinate the source and target field data to improve performance on multiple target capabilities.*

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**S6.** Explicitly stage-aware data selection for post-training was published on arXiv on 3 June 2026 (He, Zhang, Wang, Li, arXiv:2606.04466, cs.CL), which partially occupies the project's Gap 1 novelty niche: the paper organises a single training pool into distinct per-stage sets for SFT versus RL, and its own related-work framing asserts that this had not been systematically done before.

Source: <https://arxiv.org/abs/2606.04466> (primary) · refuted by 0 of 3 verifiers

*Post-training Small Language Models (SLMs) for reasoning typically follows an SFT-then-RL pipeline, yet existing work rarely considers what data should be learned at each stage.*

---

**S7.** The paper does not close Gap 2 (multi-objective or multi-capability selection) and does not touch continual pretraining. Its pipeline is SFT-then-RL only, its models are Qwen2.5-0.5B-Instruct and Llama3.2-1B-Instruct, and its five benchmarks (GSM8K-Platinum, MAWPS, SVAMP, MATH500, LogiQA) all measure a single capability, mathematical and logical reasoning, scored as aggregate accuracy. No CPT stage, no medical domain, and no multiple preserved capabilities appear anywhere in the work.

Source: <https://arxiv.org/abs/2606.04466> (primary) · refuted by 0 of 3 verifiers

*Our results highlight the importance of coordinating data difficulty across SFT and RL for effective SLM reasoning post-training.*

---

**S8.** The paper states the same load-bearing premise as the project under study, that different training stages want different data because they are credited with different learning roles, and instantiates it concretely as a difficulty split: medium-difficulty (correctness rate roughly 0.25 to 0.75) samples plus Bridge-rewritten hard samples go to SFT, while only non-all-zero-reward samples go to RL. (The 0.25 to 0.75 threshold and the per-stage set composition come from the paper body read via automated fetch of the arXiv HTML, not from the abstract, so verify against the PDF before citing.)

Source: <https://arxiv.org/abs/2606.04466> (primary) · refuted by 0 of 3 verifiers

*We argue that data strategy should be aligned with the distinct roles of SFT and RL: SFT is better suited for acquiring not-yet-mastered reasoning skills, while RL is better suited for consolidating skills that the model can already partially access.*

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**S9.** The KDD 2026 paper "Holistic Data Scheduler" (Dang, Ma, Liao; arXiv:2606.24133) already publishes something branded "multi-objective" data selection, but its three objectives are internal training signals (data quality score, inter-domain loss influence, model weight norms) rather than multiple downstream capabilities, so it does NOT close the project's Gap 2 as the project defines it (preserving several distinct clinical capabilities instead of one aggregate metric). A reviewer will nonetheless cite it as nearest prior art on the phrase "multi-objective data selection", so the review must distinguish multi-objective REWARD from multi-capability TARGET explicitly.

Source: <https://arxiv.org/abs/2606.24133> (primary) · refuted by 0 of 3 verifiers

*At the core of HDS lies a novel multi-objective, holistic reward function that integrates three critical perspectives: a data-driven reward for quality, a loss-driven reward capturing inter-domain influence, and a model-driven reward based on weight norms.*

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**S10.** HDS is scoped entirely to the pre-training stage, with no supervised fine-tuning or preference-optimisation stage and no cross-stage policy transfer, so this paper leaves Gap 1 (stage-specific selection policies across CPT, SFT, and DPO) open as of June 2026. It is also from-scratch pre-training on The Pile, not domain-adaptive continual pretraining, so it does not address whether an online mixing policy transfers to CPT of an already-pretrained base model such as Gemma 3.

Source: <https://arxiv.org/abs/2606.24133> (primary) · refuted by 0 of 3 verifiers

*The composition of training data, governed by the diversity of sources and their mixing strategy, is a cornerstone of Large Language Model (LLM) pre-training. Online Data Mixing (ODM), the technique of adaptively adjusting data mixtures during training, has emerged as a promising direction to improve efficiency.*

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**S11.** Moving the small domain dataset that is conventionally reserved for finetuning into the pretraining mixture ("specialized pretraining", SPT) cuts the pretraining tokens required to reach a fixed domain performance level by as much as 1.75x, measured across ChemPile, MusicPile and ProofPile. NOTE: figure taken from the abstract, not verified against the paper body; the domains tested are chemistry, music and formal proofs, NOT medicine, so transfer to a medical CPT corpus is untested.

Source: <https://arxiv.org/abs/2603.16177> (primary) · refuted by 0 of 3 verifiers

*Across three specialized domains (ChemPile, MusicPile, and ProofPile), SPT improves domain performance and preserves general capabilities after finetuning compared to standard pretraining. In our experiments, SPT reduces the pretraining tokens needed to reach a given domain performance by up to 1.75x.*

**S12.** Meta-rater is scoped exclusively to a single training stage, from-scratch pre-training. The paper never evaluates continual pre-training, supervised fine-tuning, or preference optimisation, and never asks whether the best pre-training subset depends on downstream SFT data. It therefore does NOT close Gap 1 (single-stage scope): a 2025 ACL Best Theme Paper on multi-dimensional selection still treats selection as a one-shot, one-stage decision. Verified from the title, abstract, and a body fetch that found no discussion of SFT, DPO, instruction tuning, continual pre-training, or domain adaptation.

Source: <https://arxiv.org/abs/2504.14194> (primary) · refuted by 0 of 3 verifiers

*offering a scalable paradigm for enhancing pre-training efficiency and model capability*

**S13.** Classifier-based Quality Filtering (CQF) applied to a raw web corpus can beat training directly on the high-quality reference set itself, even when that high-quality set is large enough not to be the binding constraint. The paper reports 53.8% for the CQF-filtered model versus 50.1% for the model trained on the unlimited high-quality set under identical training conditions (350M parameter models; the fetched context attributes this comparison to OpenOrca as the HQ set evaluated on ARC-Easy, and that dataset/benchmark attribution is lower confidence than the two numbers themselves, which are quoted from the paper body). This complicates any selection policy whose objective is 'retrieve corpus documents that look like the target domain data', which is the operating assumption behind DSIR-style distribution matching and behind naive medical-similarity scoring for CPT.

Source: <https://arxiv.org/abs/2510.00866> (primary) · refuted by 0 of 2 verifiers

*We compare it to a model trained directly on  $D_{HQ}$  under the exact same training conditions. Empirically, we observe that the model trained on the unlimited HQ set achieves an accuracy of 50.1%, while the model trained on the CQF-filtered dataset achieves 53.8%.*

**S14.** The optimal selection ratio is not a constant but a function of training compute: the paper fits  $F_{opt}(C) = 4 \times 10^{-5} \times C^{0.25}$ , meaning the optimal fraction of the corpus kept rises from about 3 percent at  $1e20$  FLOPs to about 30 percent at  $1e23$  FLOPs. Larger models require LESS aggressive filtering. This is falsifiable and directly load-bearing for a CPT-stage selection policy: a selection ratio tuned at one compute budget is provably wrong at another, so any single fixed keep-rate reported as 'the' policy is a scale-specific artefact. It also implies a small-compute project should filter far more aggressively than published large-scale recipes suggest. (Numbers from the arXiv HTML v1 body, not the abstract.)

Source: <https://arxiv.org/abs/2507.12466> (primary) · refuted by 0 of 3 verifiers

*Optimal filtering rate depends strongly on scale ...  $F_{opt}(C)=4 \times 10^{-5} \times C^{0.25}$  ... from 3% at  $10^{20}$  FLOPs to 30% at  $10^{23}$  FLOPs.*

**S15.** BETR, a cheap corpus-scale selection method that ranks documents by embedding similarity to benchmark training examples and then distills that ranking into a fastText classifier, beats DCLM-Baseline (a strong quality-classifier plus dedup baseline) by a 2.1x compute multiplier and beats unfiltered data by 4.7x, improving 9 of 10 tasks at all scales. This is direct evidence that data selection at pretraining scale does beat a dedup-plus-quality-filter baseline, not just random. Evidence base is over 500 models from 1e19 to 1e22 FLOPs with fitted scaling laws. NOTE (version discrepancy, flag in review): the arXiv HTML v1 body reports 1.8x over DCLM-Baseline on DCLM-RefinedWeb and 2.5x over DCLM-Baseline on Nemotron-CC, so the 2.1x headline appears to be a later-version or pooled figure. Cite the corpus-specific body numbers, not just the abstract.

Source: <https://arxiv.org/abs/2507.12466> (primary) · refuted by 1 of 3 verifiers

*we propose benchmark-targeted ranking (BETR), a simple method that selects pretraining documents based on similarity to benchmark training examples. BETR embeds benchmark examples and a sample of pretraining documents in a shared space, scores this sample by similarity to benchmarks, then trains a lightweight classifier to predict these scores for the full corpus. We compare data selection methods by training over 500 models spanning  $10^{19}$  to  $10^{22}$  FLOPs and fitting scaling laws to them. From this, we find that simply aligning pretraining data to evaluation benchmarks using BETR achieves a 2.1x compute multiplier over DCLM-Baseline (4.7x over unfiltered data) and improves performance on 9 out of 10 tasks across all scales.*

**S16.** Below a model-size threshold the crossing point never occurs at all: the 15M model was trained to 100B total tokens and the unfiltered pool still never overtook four of the five filters, and for the 80M model the crossing point disappears entirely once the pool grows to 10B tokens. The number of epochs needed for no-filter to win also grows super-linearly with pool size at fixed 1B model size (roughly 1 epoch at 670M tokens, roughly 3 at 2B tokens, roughly 10 at 10B tokens), so the effect gets harder to reach as pools grow.

Source: <https://arxiv.org/abs/2605.19407> (primary) · refuted by 1 of 3 verifiers

*On the leftmost plot with the 80M model, we can clearly see that crossing points cease to exist, even across our evaluated pool sizes: while there is a crossing point for the smallest 670M-token pool, there is no longer a crossing point on the largest 10B-token pool as indicated by the dark orange marker.*

## Claims killed by verification (9)

These did not survive. They are recorded so nobody re-derives them.

**K1.** On a 670M-token Common Crawl pool, a 1B-parameter model trained on the UNFILTERED pool reaches a lower average validation loss (3.37 avg NLL over C4, Cosmopedia, FineWeb-Edu) than all five filtered variants tested, including RefinedWeb (3.93) and DCLM-Baseline (5.29), which retain 13% and 2.1% of the pool respectively. This is a direct corpus-scale result that selection does NOT beat the no-selection baseline once model size and step count are large enough. (Loss values for CC and RefinedWeb are stated in prose and repeated as figure annotations; the DCLM-Baseline value is read from a figure annotation only, so treat it as figure-derived.)

Source: <https://arxiv.org/abs/2605.19407> (primary) · refuted by 3 of 3 verifiers

**K2.** The mechanism by which quality filtering helps is removal of low-quality documents, not selection of documents that resemble the high-quality target set. The authors state that the retained data are not retained because they look like the HQ set, and that CQF does not simply select the documents most likely to have come from the HQ distribution. If this generalises, then gains attributed to sophisticated corpus-scale selection are largely reproducible by a noise-removal baseline (dedup plus a generic quality filter), which is exactly the baseline a CPT-stage selection method has to beat to justify itself.

Source: <https://arxiv.org/abs/2510.00866> (primary) · refuted by 2 of 2 verifiers

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**K3.** Downstream benchmark accuracy and language-modelling loss on the high-quality set diverge under quality filtering: CQF improves downstream task performance while not necessarily improving loss/perplexity on the high-quality set. This is direct evidence against using perplexity or loss on a target set as the selection signal or as the validation metric for a corpus-scale CPT selection policy.

Source: <https://arxiv.org/abs/2510.00866> (primary) · refuted by 3 of 3 verifiers

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**K4.** Unfiltered general web data is already competitive with manually curated medical pretraining corpora, which weakens the case that hand-curated domain corpora carry special value. Table 1 body numbers: unfiltered FineWeb-2-Medical scores 45.02 Min-Max / 45.45 WP versus NACHOS (the DrBERT corpus) at 45.57 / 40.26 and TransCorpus-bio-fr (translated PubMed, the TransBERT-bio-fr corpus) at 38.41 / 32.47. The released FineMed corpus is 21.1M documents and 19.2B words versus NACHOS at 2.4M documents and 1.3B words (Table 4).

Source: <https://arxiv.org/abs/2606.22079> (primary) · refuted by 3 of 3 verifiers

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**K5.** A medically adapted LLM performs far worse than a generic model of similar size when used to generate medical pretraining data, and rephraser quality does not track parameter count. Table 2 body numbers at the 100k scale: MedGemma-27B scores 22.22 Min-Max / 23.81 WP versus generic Gemma-4-26B-A4B at 81.67 / 71.43, and the smaller Qwen3.5-35B-A3B (97.56 / 95.24) beats the larger Qwen3.5-122B-A10B (89.30 / 73.81). This is an independent 2026 data point that medical domain adaptation is not automatically an improvement over the general model, consistent in direction with Jeong et al. (arXiv:2411.08870), though measured in a data-generation role rather than on downstream medical QA.

Source: <https://arxiv.org/abs/2606.22079> (primary) · refuted by 3 of 3 verifiers

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**K6.** Naive continual pretraining on target-field data alone raises target-domain performance by 22.77% average across four target models but drops general-benchmark performance by 11.96% average, quantifying catastrophic forgetting in CPT. Mixture-based reweighting recovers 11.99% of that source-domain loss at a cost of only 0.57% on the target domain. Numbers are from the paper body (Appendix E results discussion). This is a concrete capability-retention measurement of exactly the kind angle 4 asks for, and it sets a numeric expectation for what a medical CPT run should be checked against.

Source: <https://arxiv.org/abs/2507.15640> (primary) · refuted by 2 of 3 verifiers

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**K7.** The paper explicitly characterises the published state of the art in online data mixing as of mid-2026 as single-perspective optimisation, which corroborates the project's Gap 2 premise (that existing selection is optimised against one objective) coming from an independent, non-medical group. This is useful as a citable third-party statement of the gap, but it also shows the gap is now being actively contested rather than untouched.

Source: <https://arxiv.org/abs/2606.24133> (primary) · refuted by 3 of 3 verifiers

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**K8.** The paper's normative conclusion is that domain data should be pushed as early in training as possible, which is a direct challenge to the project's framing that CPT, SFT and DPO consume categorically different data requiring separate per-stage selection policies. It argues the stage boundary should be blurred rather than that each stage needs its own optimised policy. This is prior art occupying part of the cross-stage data-allocation niche, and it is disconfirming for the strong version of Gap 1.

Source: <https://arxiv.org/abs/2603.16177> (primary) · refuted by 3 of 3 verifiers

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**K9.** Under domain shift away from web text, parameter scale can be traded for correct stage placement of domain data: a 1B-parameter SPT model beats a 3B-parameter standard-pretrained model on domains underrepresented in the pretraining corpus. NOTE: abstract-sourced figure; no confidence interval, benchmark name or evaluation protocol is given on the abstract page.

Source: <https://arxiv.org/abs/2603.16177> (primary) · refuted by 3 of 3 verifiers

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